



CANDIDATE GUIDE

DASCATM
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Hello!

Welcome to the starting blocks of your SDS™ journey!

This Candidate Guide seeks to share some of the key structural features of the SDS™ program, and also some important aspects of the responsibilities of data scientists - those that will help you plan for not only the SDS™ exam but also for meeting the larger expectations that the industry has from you as an SDS™ credential-holder.

To begin with, it is important to appreciate that the 7 knowledge areas prescribed by DASCA for senior data scientists are intricately connected to the real-world roles they get to play in the most complex of organizations. To most SDS™ applicants, this fact has been the strongest indicator that probably the best way to prepare for the SDS™ exam is to prepare yourself for meeting the expectations of data leadership and technical expertise that organizations have from their data scientists.

For this reason alone, before you can dive into the SDS™ exam section, we decided to take you through an overview of the skill- and knowledge expectations that the industry has from you. Needless to say, these skills and knowledge expectations have found their way into the SDS™ exam fabric too.

In the final two sections of the guide, you will get to know SDS™ exams up close. You may want to especially attend to the section in which we share information on what and how to study to be able to prepare for your exam optimally.

Best Wishes!

Skills and Knowledge Potential...

Earning the SDS™ is not about qualifying the SDS™ certification exam alone. It is also about your readiness to meet the expectations of organizations that their data scientists must be transformational, visionary and insightful. Let's read what knowledge & skill demands do these expectations pose on you.



Impact Revenues, Costs & Growth

As a **Senior Data Scientist**, you must have the knowledge, skills, and experience required for driving the strategies and operations of organizations through data intelligence. Organizations look up to their data scientists for insights that would help them do things faster, earlier, and more cost-effectively than their competitors can. SDS™ credential holders reflect the acumen to work across industries in organizations at any stage of their careers.

Advanced Skills to Enhance Efficiencies

Senior Data Scientists work in the top rung of the organizational hierarchy, often leading the business intelligence function in many organizations. They are expected to have a wide knowledge base and deep skill sets to be able to:

- Enable business functions and organizational departments for capturing and leveraging big data to improve outcomes
- Develop from scratch, or overhaul, the data management infrastructure in organizations
- Lead internal DevOps teams or external developers in building and commissioning data analytics and business intelligence applications and modules
- Empower department heads with the right set of functional knowledge and insights so that they can become agile and proactive
- Help develop a robust data ownership and data governance fabric within the organization.

Perspective for Smart Change

Successful and high-impact **Senior Data Scientists** are also known for their sharp perspective on functional scorecards, success metrics, business strategies, and requirements of operational excellence. As a senior data professional vying for the SDS™ credential, you must display an eye for opportunities to win:

- Linking managerial performance with improved usage of big data
- Rapidly executing data strategies for various departments
- Overcoming legacy disadvantages
- Building an optimal combination of technologies, teams, and models
- Leveraging data to ride out of storms and to deliver competitive superiority to the organization

Vision for Business Intelligence

As a **Senior Data Scientist** and a data leader, you are expected to transform the very way organizations can deploy data to generate intelligence that breaks new ground for them and keeps them ahead. Data Scientists enable organizations to become nimble, agile, and proactive during times when change is comprehensive, rapid, and unpredictable, and impacts all functions and departments. Some key areas of focus for you must be:

- Dramatically improving outcomes of systems and processes by automating them and driving them through precise, smart data, rather than fueling them with discretionary manual inputs.
- Equipping organizational strategists with intelligence on the most lucrative new markets, product segments, and business opportunities
- Empowering departmental leaders and managers with the mechanisms through which they can tap intelligence that speeds up decision-making; optimizes outcomes and reduces errors of omission.

About SDS™ exam

The SDS™ exam carries multiple-choice, single-answer questions carrying equal marks, with no penal scoring for wrong answers. A typical DASCAs certification exam carries a mix of questions that seek to test the accuracy of knowledge of principles and concepts; understanding of techniques and processes; and the skills to apply tools and techniques of programming and analysis in solving real data problems. The exam questions are drawn from 7 knowledge areas prescribed for data scientists in the DASCAs framework.

Exam Coverage

Data Science Applications in Business	15%
Framework of data science use-cases for business; organizational strategy and data analytics; information, intelligence and visualization approaches of data scientists; "by" analysis; score development; monetization and metamorphosis exercises.	
The Functional Framework of Data Science	18%
The framework of data science functions and tools; interconnection of business intelligence, big data analytics and data science; the Data Lake; applications of analytics in organizational decision-making and strategy implementation.	
Driving Business Through Big Data Analytics	17%
The big data business mandate; Big data business model maturity index; preparing the big data strategy document; the importance of the user experience; catalyzing business performance and outcomes through business intelligence.	
Building Pan-Organization Support	15%
Developing cross-departmental buy-ins to analytics-led functioning; envisioning data science impacts on organizational performance; leading development of enterprise-wide data analytics ecosystem.	
Data Science Essentials for Data Scientists	15%
The Data science roadmap; programming languages; visualization and simple metrics; machine learning fundamentals; machine learning classification; technical communication and documentation.	
Big Data Analytics Essentials for Data Scientists	10%
Unsupervised learning: clustering and dimensionality reduction; regression fundamentals; Data Encoding and File Formats; High-volume streaming data storage; real-time streaming data architectures; stream and batch data processing; software engineering best practices; Natural Language Processing; Time Series Analysis; fundamentals of Statistics, Linear Algebra and Probability.	
Advanced Data Science for Data Scientists	10%
Performance and computer memory; computer memory and data structures; maximum likelihood estimation and optimization; advanced classifiers; stochastic modelling.	

Exam Format

The SDS™ examination aims to check and validate the accuracy and breadth of the candidate's core knowledge and skills related to advanced data analytics; data governance; business intelligence infrastructure among other topics included in the knowledge areas.

Know The Question Format

SDS™ exam questions may fall within any of the following three categories:

TYPE A- GENERAL UNDERSTANDING OF DATA SCIENCE

Type A questions directly test the candidates' ability to correctly and precisely recollect concepts, principles, techniques, and generally accepted practices in Data Science. The answers to TYPE A questions will reflect candidates' awareness and comprehension of critical Data Science concepts elements and issues as enshrined in the DASCA Body of Knowledge. There is only one correct answer to TYPE A questions, candidates are scored only if the correct option is selected.

TYPE B- ADVANCED UNDERSTANDING OF DATA SCIENCE

Type B questions test the candidates' understanding of Data Science at an advanced level of Data Science techniques and practices. TYPE B Questions would be fact-based with answer options being clear statements – either clearly correct, or incorrect.

TYPE C- IMPLEMENTATION OF DATA SCIENCE TECHNOLOGIES

Type C questions directly test the candidates' ability to APPLY and DEPLOY their awareness and understanding of the Data Science technologies in real-life practice. There is only one correct answer to TYPE C questions, candidates are scored only if the correct option is selected.

Examples

TYPE A - GENERAL UNDERSTANDING OF DATA SCIENCE

Question 1:

The first Fraud Management System was based on

- a) CPP b) NPP c) MPP d) RPP

Correct Answer: c) MPP

Question 2:

HBase does not support which of the following?

- a) Compaction b) MapReduce jobs c) Hashing d) Secondary indexes

Correct Answer: d) Secondary indexes

TYPE B - ADVANCED UNDERSTANDING OF DATA SCIENCE

Question 1:

What could be the purpose of an online retailer's analytics team using 'Collaborative filtering' algorithm?

- a) Recommendation of the product b) Shelf space allocation
c) Shopper navigation control d) Pricing optimization

Correct Answer: (a) Recommendation of the product

Question 2:

An analyst has obtained data from the following sources. Identify the source where Hive CANNOT be used to analyze the data.

- a) Twitter data b) Web logs c) Call data records d) Oracle CRM tables

Correct Answer: d) Oracle CRM tables

TYPE C - APPLIED & PRACTICAL UNDERSTANDING OF DATA SCIENCE

Question 1:

Which of the following commands is usually the best for dropping database, along with the tables in Hive?

- a) DROP DATABASE my_database ALL b) DROP DATABASE my_database FALSE
c) DROP DATABASE my_database TRUE d) DROP DATABASE my_database CASCADE

Correct Answer: d) DROP DATABASE my_database CASCADE

Question 2:

What could be the purpose of an online retailer's analytics team using 'Collaborative filtering' algorithm?

- a) Recommendation of the product b) Shelf space allocation
c) Shopper navigation control d) Pricing optimization

Correct Answer: a) Recommendation of the product

What and How to Study for the SDS™ Exam

The SDS™ exam carries multi-choice-Single-answer, The SDS™ exam features 85 multiple-choice questions with only one correct answer to be completed in 100 minutes in an online and remote-proctored setting. As you may have already noted from the previous sections, these questions cover 7 topic clusters, carrying various degrees of weightage. Given past data and candidate experience, we can suggest some key aspects to focus on while preparing for these 7 topic clusters.

Data Science Applications in Business

This cluster tests your general understanding of how to deploy big data analytics and business intelligence systems to impact the strategic, operational, and competitive capabilities of an organization. More specifically, you must focus on understanding the following:

1. What kind of data science is valuable for businesses: large-data processing and analysis, predictive capabilities, information-distillation, machine learning and automation, intelligence, and visualization, "by" analysis, score development, monetization, and metamorphosis exercises.
2. Strategic use-cases for data science and data analytics' role in impacting business competitiveness, survival, and growth: how does big data analytics help firms develop better strategic options for growth, crises management, new product development, and new projects?
3. Where data science and data analytics tangibly impacts business operations: how can big data analytics help sharpen marketing, sales, production, HR, research, investing, supply chain, and operational costs?
4. The critical technologies deployed for data analytics/ business intelligence systems in organizations today.

The Functional Framework of Data Science

This cluster tests your general understanding of the functional and operational framework of data science as it is deployed in organizations across industries and sectors. This cluster includes functions and tools, interconnections of business intelligence, big data analytics, and data science, management of data lakes, applications of analytics in organizational decision-making, and strategy implementation. More specifically, you must focus on having thorough knowledge about:

1. The types of decisions data scientists have to take in routine and non-routine work. These decisions include developmental and crises-managing, advising the board and functional heads, pure-play technology project management, and infrastructure creation or overhauling.
2. Ways in which the data science/ data analytics/ business intelligence department or unit can be structured in an organization to ensure that data analytics and business intelligence management are integrated laterally and drive managerial activity across functions and departments.
3. The structure and dynamics of how functional information systems and legacy software connect to business intelligence infrastructure, and what are the key attention areas, showstoppers, and best practices.
4. Typical data scientist functions in modern organizations – ranging from the development of analytics infrastructure to imagining and executing data capture and distribution strategies, to management of policies and processes of data security, data governance and ownership, and creating a thriving cross-functional business intelligence fabric.
5. Modern critical technologies deployed in various decision areas and functions in the data analytics/business intelligence departments/units in organization.

Driving Business through Big Data Analytics

This cluster tests your understanding of established and evolving big data business mandates in organizations. You must enrich your perspective on how data science drives transformative big data analytics and fuels adaptability, efficiency, and operational security in organizations. This includes, but is not limited to the following concepts:

1. Applications of the big data business model maturity index and being familiar with cases of how organizations in different stages of their life cycles and legacy maturity can leverage data intelligence strategies.
2. The various approaches, rubrics, and processes involved in creating top-level big data strategy documentation for an organization, and the constituents, structure, and alternative formats of such documents.
3. The importance of user experience in the success of organizational data strategies and the challenges typically experienced by different functional departments when they embrace a data-led approach to enhancing user engagement.
4. The specific business intelligence that is needed by different departments to catalyze their performance and drive significantly improved outcomes for them.

Building Pan-Organization Support

This cluster tests your general understanding of how to develop cross-departmental buy-ins to analytics that drives a positive impact on organizational performance through the development of an enterprise-wide data analytics ecosystem. More specifically, you must focus on having a solid foundational understanding in:

1. The issues and challenges of generating buy-ins among managers for analytics-led functioning and how this has been done successfully in other organizations. The focus must be on learning the leadership and managerial processes and approaches to generating buy-ins rapidly.
2. Most important desired outcomes for each data science department and the possibilities of using data lakes and data intelligence to transform the performance of these departments.
3. The ideas and processes you can deploy while engaging the talent of other departments to contribute to the evolution of a robust enterprise-wide data infrastructure.
4. Data literacy and how to develop and execute smart strategies, policies, plans, and processes to empower departments across the board with the consciousness as well as functional knowledge of data analytics.
5. Managing Data Governance in organizations and how to develop and execute relevant policies, processes, and standards associated with data assets.

Data Science Essentials for Data Scientists

This cluster tests your deeper technical knowledge related to road-mapping analytics development in organizations; programming languages; visualization and simple metrics; machine learning fundamentals and applications and technical communication and documentation. The most important skills in this category include:

1. Building a strong mental framework of alternative approaches to developing data science roadmaps for organizations, and how this is done in the organization at various stages of life cycles with diverse legacy infrastructure.
2. Working knowledge of database basics, relational algebra, matrices & linear algebra functions, extract transform load, and hash functions & binary trees.
3. Understanding the differences between data science, data engineering, business analytics, ML, and deep learning, and the concepts of unsupervised and supervised learning.
4. Foundational applied knowledge of developing data models and analytical models. Any function in data science requires programming, so knowing either Python, R, Java, or SQL will be very important.
5. Familiarity with the software engineering process. Data science often involves software logging and the development of data-driven products. Skills in software engineering will come in handy in all these tasks.
6. Confidence in utilizing data visualization strategies and models that organizational departments and top management typically need.

Big Data Analytics Essentials for Data Scientists

This cluster tests your in-depth technical knowledge about data encoding and file formats, high-volume streaming data storage, real-time streaming data architectures, stream and batch data processing, software engineering best practices, natural language processing, time series analysis, fundamentals of statistics, linear algebra and probability, unsupervised learning, clustering and dimensionality reduction, and regression fundamentals. More specifically, you must focus on the following:

1. Familiarity with data manipulation and data wrangling and at least one Data Manipulation Language (DML) to organize, arrange, and change data to make it readable.
2. Navigating vast amounts of data without missing important details through the ability to spot patterns, anomalies, and discrepancies, especially in unstructured data.
3. Developing data and analytical models and dealing with unstructured data using Python, Perl, C/C++, SQL, and Java.
4. Aptitude data visualization. Besides the ability to create a visualization program and standards for the entire organization to follow, you must also have hands-on skills in at least one of the leading applications like Datawrapper, Tableau, Kibana, and Google Charts.
5. Knowledge of machine learning, fundamentals of deep learning, and the basics of neural networks used to build deep learning models like Keras. You must also check again on Autoencoders, RBM, and recurrent and convolutional neural networks. organizational departments and top management typically need.

Advanced Data Science for Data Scientists

This cluster tests your knowledge about the more engaged and advanced aspects of data science applications, especially related to deploying big streaming data in automation, real-time sensing, and unsupervised learning. You will need to refresh your knowledge about the foundations of performance and computer memory, advanced data structures, maximum likelihood, estimation, optimization, and advanced classifiers and stochastic modeling. This includes:

1. Understanding of computer memory systems and distributed computing applications in data science.
2. Dealing with the more complex mathematical techniques of handling big data and analyzing big data for decision-making through statistical inference.
3. Applied knowledge about massive parallel data processing, data exploration, and visualization, advanced machine learning, deep learning and augmented learning.
4. Familiarity with machine learning & deep learning algorithms and their practical use cases, and learning how to take architectural decisions, understanding the characteristics of different algorithms, frameworks & technologies, and how they impact model performance and scalability through this information.
5. Developing an understanding of stochastic modeling of complex real-world, unstructured problems and advanced data intelligence systems using these models.

Recommendations

It is important to note that despite the growing career opportunities in data science, there is a lingering confusion among the general population on how a data analyst and technologist differs from a traditional data scientist. This lack of clarity creates unfair expectations on data scientists – who are at times expected to know what data analysts and engineers do, something that you must keep in mind as you start your professional journey toward becoming a data scientist.

The fast-paced evolution of the industry is a key issue too; little written content about data science stays relevant for too long. The following resources and publications can help you stay up to date with the latest data science developments:

- **International Journal of Data Science and Analytics**
- **Big Data**
- **Big Data Research**
- **Journal of Big Data**
- **International Journal of Business Intelligence and Data Mining**
- **Computational Statistics & Data Analysis**
- **Sigkdd Explorations**
- **Artificial Intelligence**
- **Foundations and Trends® in Machine Learning**
- **Social Network Analysis and Mining**
- **Advances in Data Analysis and Classification**

- **Intelligent Data Analysis**
- **Predictive Analytics World**
- **Towards Data Science**
- **Inside BIG DATA**
- **Data Science Central**
- **Information Week**
- **Gartner**
- **Dataflop**
- **Forrester**
- **Kdnuggets**
- **Dataconomy**
- **TechCrunch**
- **CIO Magazine**
- **DataRobot**
- **Forbes**

Exam-preparation Resources & Exam Policies

All registered SDS™ candidates receive print copies of the Data Science Handbooks. These handbooks are prescribed for exam preparation.

DASCA practices stringent examination standards to ensure the highest integrity of the exam content and fairness for all examinees during the delivery of the examination. It is the responsibility of the candidate to familiarize themselves with the policies and procedures before they appear for the examination.

DASCA official exam-preparation resources

SDS™ candidates get a pair of Data Science handbooks exclusively prepared in accordance with the DASCA knowledge framework. In addition to serving as a base of learning and exam preparation, these handbooks double up as useful reference resources for candidates even after they have earned their SDS™ credential.

Data Science Handbook 1 contains 23 chapters covering intricate details on all technical aspects of big data, data science, data analytics, and their applications.

The 15 chapters of the Data Science Handbook 2 are focused on data science applications, and straddle across 4 salient themes - Business Potential of Big Data; Data Science Overview; Data Science for Business Stakeholders, and Building Cross-Organizational Support.



In addition, DASCA partners provide online resources including videos for supplemental learning.

System Requirements

Below are the minimum system requirements to appear for the DASCA exams:

System Essentials	Requirements
Display (Resolution)	1024 x 768 [Recommended]
Network Connection	Strongly recommend using a wired network rather than a wireless network as it provides better exam delivery performance.
Internet Connection Speed	Minimum 1 Mbps Please note: Candidates need to ensure that to run the exam smoothly, their internet speed must be at least 1 MBPS. In the event their exam interrupts and they cannot complete it because of bandwidth reasons at their end, candidates can reschedule it for another slot from their myDASCA dashboard after paying a nominal fee.
Web Browser(s)	Latest Version of Google Chrome Only
Webcam	Internal or an External Webcam must be connected to the computer
Firewall(s)	Corporate Firewalls (including VPNs) often cause this Exam Delivery method to fail. Please take the exam in a setting without these firewalls
Admin Rights	Should have admin rights on your laptop/computer if you are taking the exam from your office premises.

Identification Policy

While scheduling the SDS™ exam, you will be required to submit details of two valid, nonexpired Government-issued identification documents. The IDs must be valid and current (not expired) on the exam day. One of the IDs must be a photo ID with your legal name and signature. Your legal name on the ID must exactly match the first and last name on your myDASCA account.

Both IDs must be issued by the country in which the candidate is testing. If the candidate does not have a qualifying primary ID issued from the country they are testing in, an International Travel Passport from their country of citizenship is required, along with a secondary ID.

List of acceptable IDs -

- Government-issued driver's license
- Passport
- Military ID
- State/country ID card
- Employee/school ID

You will be required to present your ID on the screen to the remote proctor on exam day. If there is any discrepancy in the ID presented or the ID does not fulfill one or more requirements from above, the remote proctor has the right to refuse entry into the exam zone on the platform.

It is also recommended that you ensure the exam room is well-lit and your webcam has a good resolution to ensure a smooth verification process.

Scheduling, Rescheduling, and Cancellation Policy

Exam Scheduling

Once you complete your registration, you have a period of 180 days to study, prepare for and give your exam. It is the responsibility of the candidate to schedule the exam in time to ensure they get access to preferable exam windows. It is recommended you schedule your examination at least 30 days before the expiry of 180 days period. Any requests for exam scheduling within 20-30 days of the expiry of the 180 days validity period will not be accommodated. To schedule your exam after the expiry of 180 days, you can do so after paying a nominal fee on your myDASCA dashboard.

In the event you do not show up for your exam as scheduled, you will be marked 'No-show'. If you miss your scheduled examination due to an emergency, please contact DASCA within 48-96 hours to report the matter. Visit the policy section [here](#) to read more about Appeals.

Exam Rescheduling

To reschedule your exam, you must log in to your myDASCA dashboard 72 hours prior to your scheduled examination. Rescheduling in less than 72 hours prior to your scheduled exam will require you to pay USD 50 from your myDASCA dashboard.

Cancellation Policy

If you wish to cancel your exam, you should log in to your myDASCA dashboard 72 hours prior to your scheduled examination. To reschedule the exam after voluntary cancellation, you can immediately reapply from your myDASCA dashboard by paying a nominal fee of USD 50.

Examination Reappear Policy

In the event you fail your first attempt to pass the examination, you can apply to reappear after your result is published on your myDASCA dashboard. You will be required to pay a nominal fee of USD 50 and then schedule the exam for a later date.

Exam Conduct

All examinees must comply with DASCA's policies and procedures to ensure a fair and secure examination process. DASCA has a zero-tolerance policy for any misconduct or adoption of unfair means in its ecosystem. All decisions taken by the remote proctor during the exam session will be final and binding including termination of the examination due to non-compliance or any activity that is deemed suspicious or not in compliance with the code of conduct. Examinees cannot appeal the final decision regarding an exam policy violation.

It is recommended that you are at your desk at least 30-50 minutes prior to your exam start time. This gives you ample time to check your internet connectivity, and system compatibility, give the mock exam and ensure you have read through and familiarized yourself with all the terms, conditions, and policies well in advance. We encourage you to review the guidelines below to gain perspective and understanding of an examinee's conduct during the examination. If you have any specific questions, please [contact us](#) to get more clarity before you appear for your examination. No personal belongings are allowed in the exam room including mobile phones, digital assistants, reading material including the DASCA handbooks, tablets, e-readers, digital watches, and any other electronic device except the computer/ laptop you will be appearing for the exam on. You are only allowed to have a webcam and an external mouse connected to your device.

No breaks are allowed during the examination. You should be visible at all times on the exam screen. If you are partially visible on the exam screen at any time during the exam, your exam may be terminated. Trying to relocate during the examination is disallowed as well.

Make sure your browser tabs and applications are closed/ signed out during the exam, especially if you are using a work laptop/ computer. Any notification pop-ups of email and chat will lead to the immediate termination of the exam session.

Exam Integrity

The examination and all connected materials are the intellectual property of the Data Science Council of America. Examinees should not try to copy the exam questions or take a screenshot or record during the examination through any means. After the exam has been administered, the examinees are prohibited to publish or share the examination content in any format. Any such act of reproducing or distributing content or any material, that violates the copyright laws, may invite stringent penal action. Any action, or activity by the examinee including any misconduct, intentional or otherwise, during or after the administration of the exam, will be deemed as a violation of DASCA's policies and procedures.



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